higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

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This marking guideline consists of 13 pages.

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QUESTION 1

1.1 1.1.1

$$y = \lim_{x\to 3}\left[\frac{1}{\ln(x-2)} - \frac{1}{(x-3)}\right]$$

$$= \lim_{x\to 3}\left[\frac{(x-3)-\ln(x-2)}{(x-3)\ln(x-2)}\right] \checkmark \quad \frac{0}{0}$$

$$= \lim_{x\to 3}\left[\frac{1-\frac{1}{x-2}}{\ln(x-2)+(x-3)\times\frac{1}{x-2}}\right] \checkmark \quad \frac{0}{0}$$

$$\lim_{x\to 3}\left[\frac{\frac{1}{(x-2)^2}}{\frac{1}{x-2}+\frac{(x-2)-(x-3)}{(x-2)^2}}\right] \checkmark$$

$$= \frac{1}{2} \checkmark$$ (4)

1.1.2

$$y = \lim_{x\to 0}\frac{\log(x+1)-\sin x}{3x} \quad \frac{0}{0}$$

$$= \lim_{x\to 0}\frac{\frac{1}{\ln 10}\times\frac{1}{x+1}-\cos x}{3} \checkmark$$

$$= \frac{\frac{1}{\ln 10}-1}{3} \checkmark$$

$$= -0{,}189 \checkmark$$ (3)

1.2

$$y = \frac{x^3+5x-10}{x^2-3x+2}$$

$$x^2-3x+2=0$$

$$x = 1\checkmark \quad or \quad x = 2\checkmark$$

(2)
[6]

QUESTION 2

2.1 $f(x) = \dfrac{x-10}{x+2}$

$f'(x+h) = \lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$

$\dfrac{dy}{dx} = \lim\limits_{h\to 0} \dfrac{\frac{(x+h)-10}{(x+h)+2} - \frac{x-10}{x+2}}{h}$ ✓

$= \lim\limits_{h\to 0} \dfrac{\frac{x^2+2x+xh+2h-10x-20-(x^2-10x+xh-10h-2x+20)}{(x+2)(x+h+2)}}{h}$ ✓

$= \lim\limits_{h\to 0} \dfrac{\frac{12h}{(x+2)(x+h+2)}}{h}$ ✓

$= \lim\limits_{h\to 0} \dfrac{12}{(x+2)(x+h+2)}$

$= -\dfrac{12}{(x+2)^2}$ ✓ (4)

2.2 2.2.1 $y = \sin^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$

$\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-\left(\frac{1-x^2}{1+x^2}\right)^2}}$ ✓ $\times \left[\dfrac{-2x(1+x^2)✓ - 2x(1-x^2)✓}{(1+x^2)^2✓}\right]$

(4)

2.2.2 $y = \dfrac{e^{x\cos x}}{1+e^x}$

$\dfrac{dy}{dx} = \dfrac{(1+e^x)[e^{x\cos x}(\cos x✓ + x.-\sin x✓)] - e^{x\cos x}\times e^x✓}{(1+e^x)^2✓}$ (4)

2.3 $y = (3x - 2^x)^{\cos 3x}$

$\ln y = \cos 3x \times \ln(3x - 2^x)$

$\frac{1}{y}\frac{dy}{dx}$✓ $= \cos 3x \times \frac{1}{3x - 2^x} \times (3 - 2^x \ln 2)$✓ $- 3\sin 3x \times \ln(3x - 2^x)$✓

$\frac{dy}{dx} = y\left[\frac{(3 - 2^x)\cos 3x}{3x - 2^x} - 3\sin 3x \times \ln(3x - 2^x)\right]$✓

$= (3x - 2^x)^{\cos 3x}\left[\frac{(3 - 2^x)\cos 3x}{3x - 2^x} - 3\sin 3x \times \ln(3x - 2^x)\right]$ (4)

2.4 2.4.1 $\frac{9 - x^2}{(y - x)^3} = 1$

$9 - x^2 = (y - x)^3$

$0 - 2x$✓ $= 3(y - x)^2 \times \left(\frac{dy}{dx} - 1\right)$✓

$\frac{dy}{dx} - 1 = -\frac{2x}{3(y - x)^2}$

$\frac{dy}{dx} = 1 - \frac{2x}{3(y - x)^2}$✓ (3)

2.4.2 $\frac{dy}{dx} = 1 - \frac{2(1)}{3(1 - 3)^2} = \frac{5}{6}$✓

$y = \frac{5}{6}x + c$

$3 = \frac{5}{6}(1) + c$

$c = \frac{13}{6}$✓

$y = \frac{5}{6}x + \frac{13}{6}$✓ (3)

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QUESTION 3

3.1 3.1.1 $f(x) = x^3 - 2x^2 - 3x + 5$

$f'(x) = 3x^2 - 4x - 3 = 0$✓

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-3)}}{2(3)}$$

$x = -0{,}535 \ \ or \ \ x = 1{,}869$

$y = 7{,}024 \quad y = -1{,}065$

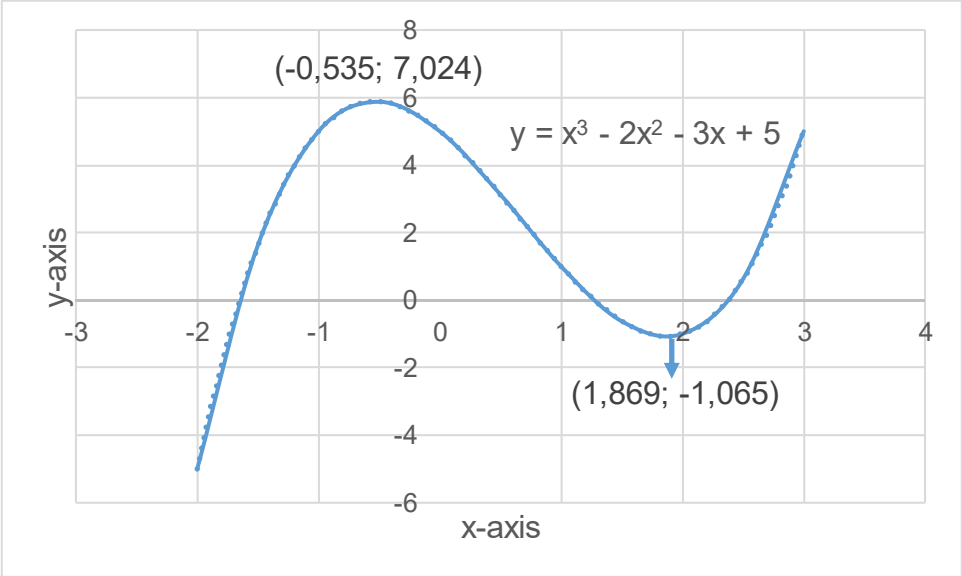
Thus, the coordinates of the turning points are $(-0{,}535 \ ; \ 7{,}024)$✓ and $(1{,}869; -1{,}065)$✓ (3)

3.1.2

x	-2	-1	0	1	2	3
y	-5	5	5	1	-1	5

✓= 1 Mark for any 3 correct answers (2)

3.1.3



✓ = 1 Mark for the shape
✓= 1 Mark for the indication of the turning point on the graph (2)

3.1.4 $Let\ x_0 = 1{,}3$

$f(1{,}3) = -0{,}083$

$f'(1{,}3) = -3{,}13$✓

$x_1 = 1{,}3 - \frac{-0{,}083}{-3{,}13}$✓

$= 1{,}273$✓

$f(1{,}273) = 0{,}003$

$f'(1{,}273) = -3{,}230$

$x_2 = 1{,}273 - \frac{0{,}003}{-3{,}230}$

$= 1{,}274$✓ (4)

3.2 $Perimeter = 2l + w$

$160 = 2l + w$

$w = 160 - 2l$✓

$Area = w \times l$

$A = l(160 - 2l)$

$= 160l - 2l^2$✓

$A' = 160 - 4l = 0$✓

$l = 40\ m$✓

$w = 160 - 2 \times 40 = 80\ m$✓ (5)

3.3 $V = \frac{4}{3}\pi r^3$

$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$✓

$7\text{cm}^3/\text{s} = 4\pi(4\text{ cm})^2 \frac{dr}{dt}$✓

$\frac{dr}{dt} = \frac{7\text{cm}^3/\text{s}}{4\pi(4\text{ cm})^2}$✓

$= 0{,}035\text{ cm/s}$✓ (4)

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QUESTION 4

4.1 4.1.1 $\int \frac{e^{\sqrt{\sin x}}}{\sec x.\sqrt{\sin x}}dx$

$let\ u = \sqrt{\sin x}$

$du = \frac{1}{2\sqrt{\sin x}}.\cos x\ dx$✓

$2\ du = \frac{1}{\sqrt{\sin x}}.\cos x\ dx$

$= 2\int e^u\ du$✓

$= 2e^u + C$

$= 2e^{\sqrt{\sin x}} + C$✓

(3)

4.1.2 $\int \sin 3x \sin(-2x)\ dx$

$= \frac{1}{2}\int (\sin x + \sin 5x)\ dx$✓

$= \frac{1}{2}\left(-cos\ x - \frac{\cos 5x}{5}\right) + C$

$= -\frac{\cos x}{2}$✓ $-\frac{\cos 5x}{5}$✓ $+ C$

(3)

4.1.3 $\int \cos^2 x \sin^3 x \, dx$

$= \int \sin x \sin^2 x \cos^2 x \, dx$ ✓

$let\ u = \cos x$ $du = -\sin x$ ✓

$= -\int (1-u^2)u^2 du$ ✓

$= \int (u^4 - u^2) du$

$= \frac{1}{5}u^5 - \frac{1}{3}u^3 + C$

$= \frac{1}{5}\cos^5 x$ ✓ $- \frac{1}{3}\cos^3 x$ ✓ $+ C$ (5)

4.1.4 $\int \frac{3x^3 - 2x^2 + x - 5}{x+2} dx$

$= \int \left(3x^2 - 8x + 17 - \frac{39}{x+2}\right) dx$ ✓

$= x^3$ ✓ $- 4x^2$ ✓ $+ 17x$ ✓ $- 39\ln(x+2)$ ✓ $+ C$ (5)

4.1.5 $\int \frac{x^3 e^{x^2}}{(x^2+1)^2} dx$

$let\ u = x^2 e^{x^2}$ $\qquad dv = \frac{x}{(x^2+1)^2}$ $du = \left(2x^3 e^{x^2} + 2x e^{x^2}\right) dx$ $\qquad v = -\frac{1}{2(x^2+1)}$ ✓

$\int \frac{x^3 e^{x^2}}{(x^2+1)^2} dx$

$= -\frac{1}{2(x^2+1)} \times x^2 e^{x^2} - \int -\frac{1}{2(x^2+1)} \times \left(2x^3 e^{x^2} + 2x e^{x^2}\right) dx$ ✓

$= -\frac{x^2 e^{x^2}}{2(x^2+1)} + \int \frac{2x e^{x^2}(x^2+1)}{2(x^2+1)} dx$

$= -\frac{x^2 e^{x^2}}{2(x^2+1)} + \int x\, e^{x^2} dx$ ✓

$= -\frac{x^2 e^{x^2}}{2(x^2+1)} + \frac{1}{2}e^{x^2} + C$ ✓ (4)

4.2 $\int \frac{11-3x}{x^2+2x-3}\,dx$

$\frac{11-3x}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3}$

$11-3x = A(x+3) + B(x-1)$✓

$let\ x=-3;\ \ B=-5$✓ or $x=1;\qquad A=2$✓

$\int \left(\frac{2}{x-1} + \frac{-5}{x+3}\right)dx$

$= 2\ln(x-1)$✓ $- 5\ln(x+3)$✓ $+ C$ (5)

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QUESTION 5

5.1 $I = \int_0^{\frac{\pi}{4}} \frac{\sec x \tan x}{1+\sec^2 x}\,dx$

$let\ u = \sec x$ $du = \sec x \tan x\,dx$ ✓ $x = 0 \leftrightarrow u = 1$ $x = \frac{\pi}{4} \leftrightarrow u = \sqrt{2}$ ✓

$I = \int_1^{\sqrt{2}} \frac{1}{1+u^2}\,du$ ✓

$= [tan^{-1}\,u]_1^{\sqrt{2}}$✓

$= 0{,}170$✓ (5)

5.2 $\mathcal{L}(3) = \frac{3}{s}$ ✓ (1)

5.3 5.3.1

$$A = \int_a^b (y_1 - y_2)\,dx$$

$$= \int_0^2 \left[\left(\sqrt{4-x^2}\right) - \left(\frac{1}{2}x^2 - 2x + 2\right)\right] dx$$

$$= \int_0^2 \left(\sqrt{4-x^2} - \frac{1}{2}x^2 + 2x - 2\right) dx \checkmark$$

$$= \left[2\sin^{-1}\left(\frac{x}{2}\right) - \frac{1}{6}x^3 + x^2 - 2x\right]_0^2 \checkmark$$

$$= \pi - \frac{4}{3}$$

$$= 1{,}808\ units^2 \quad \checkmark \qquad (3)$$

5.3.2

$$V = \pi \int_a^b (y_1{}^2 - y_2{}^2)\,dx$$

$$= \pi \int_0^2 \left[\left(\sqrt{4-x^2}\right)^2 - \left(\tfrac{1}{2}x^2 - 2x + 2\right)^2\right] dx \checkmark$$

$$= \pi \int_0^2 \left(4 - x^2 - \frac{1}{4}x^4 + 2x^3 - 6x^2 + 8x - 4\right) dx \checkmark$$

$$= \pi \int_0^2 \left(-\frac{1}{4}x^4 + 2x^3 - 7x^2 + 8x\right) dx \checkmark$$

$$= \pi \left[-\frac{1}{20}x^5 + \frac{1}{2}x^4 - \frac{7}{3}x^3 + 4x^2\right]_0^2 \checkmark$$

$$= \frac{56\pi}{15} = 11{,}729\ units^3 \checkmark \qquad (5)$$

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5.4 $m = \sigma A \qquad dm = \sigma dA$

$dI_z = r^2 dm$

$= x^2 \sigma dm$✓

$= x^2 \sigma 2\pi x \, dx$✓

$= 2\pi\sigma \int_0^4 x^3 \, dx$ ✓

$= 2\pi\sigma \left[\frac{1}{4}x^4\right]_0^4$ ✓

$= 2\pi\sigma \left[\frac{4^4}{4} - 0\right] \qquad = 128\pi\sigma \ \ or\ 402{,}124\ units^4$✓

(5)
[19]

QUESTION 6

6.1 $\frac{dy}{dx} = 2x - 1 + 2xy - y$, if $y(0) = 2$

$\frac{dy}{dx} = (2x - 1)(y + 1)$

$\frac{dy}{y+1} = (2x - 1)\, dx$✓

$\ln(y + 1)$✓ $= x^2 - x + C$ ✓

(3)

6.2 $\frac{1}{(x+3)^2-2}\frac{d^2y}{dx^2}=1,\quad \text{if } \frac{dy}{dx}_{|\ x=1}=0 \text{ and } y(1)=2$

$\frac{1}{(x+3)^2-2}\frac{d^2y}{dx^2}=1$

$\frac{d^2y}{dx^2}=(x+3)^2-2$

$\frac{d^2y}{dx^2}=x^2+6x+7$

$\frac{dy}{dx}=\frac{1}{3}x^3+3x^2+7x+A$✓

$0=\frac{1}{3}(1)^3+3(1)^2+7(1)+A$

$A=-\frac{31}{3}$✓

$\frac{dy}{dx}=\frac{1}{3}x^3+3x^2+7x-\frac{31}{3}$

$y=\frac{1}{12}x^4+x^3+\frac{7}{2}x^2-\frac{31}{3}x+B$✓

$2=\frac{1}{12}(1)^4-(1)^3+\frac{7}{2}(1)^2-\frac{31}{3}(1)+B$

$B=\frac{31}{4}$✓

$y=\frac{1}{12}x^4+x^3+\frac{7}{2}x^2-\frac{31}{3}x+\frac{31}{4}$✓ (5)

[8]

TOTAL: 100

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